

Phone QA & GPS outages

Measured development evidence · not release approval

Six durations. No missing points hidden.

120 s GPS warmup; 10 Hz ticks in [0, duration); missing estimates count as failures

Withheld phone GNSS, not independent surveyed ground truth

These development results do not establish the 90%-within-10-metre field target.

GPS gap	Within 10 m / expected	Success	Available	p90 error
10 s	285 / 2600	10.96%	377	10.21 m
20 s	183 / 4800	3.81%	342	19.50 m
30 s	408 / 6900	5.91%	457	8.44 m
40 s	232 / 8000	2.90%	445	19.75 m
50 s	130 / 10000	1.30%	190	11.43 m
90 s	91 / 14400	0.63%	164	1.51 m

Verified success is a conservative lower bound: known within 10 metres divided by every expected update. Missing references remain unknown. p90 errors describe comparable predictions only, not missing positions.

Three same-phone/day development rides with unconfirmed mounts

Duration runs have different cycle counts; matched prefixes use 90 s runs

Ticks and duration prefixes are correlated, not independent trials

Conditional errors exclude missing predictions; joint success does not

Missing references are unknown: joint success is a verified lower bound

Low errors on rare outputs do not prove long-outage tracking

Physical Location-off continuity is not a moving accuracy experiment

APK SHA-256: 410ca8d79d42780b3cb1c33c0d569c093e1216b257d9c2ccd10dc56e9a8db62b

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The same outages, examined in phases.

Cumulative prefixes of the same 90-second outages. These keep the test cohort fixed when comparing durations; they are not independent experiments.

GPS gap	Within 10 m / expected	Success	Available	p90 error
10 s	91 / 1600	5.69%	164	1.51 m
20 s	91 / 3200	2.84%	164	1.51 m
30 s	91 / 4800	1.90%	164	1.51 m
40 s	91 / 6400	1.42%	164	1.51 m
50 s	91 / 8000	1.14%	164	1.51 m
90 s	91 / 14400	0.63%	164	1.51 m

Non-overlapping intervals within those same outages:

GPS gap	Within 10 m / expected	Success	Available	p90 error
0–10 s	91 / 1600	5.69%	164	1.51 m
10–20 s	0 / 1600	0.00%	0	Unavailable
20–30 s	0 / 1600	0.00%	0	Unavailable
30–40 s	0 / 1600	0.00%	0	Unavailable
40–50 s	0 / 1600	0.00%	0	Unavailable
50–90 s	0 / 6400	0.00%	0	Unavailable

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The actual Android Location switch.

No independent moving reference; continuity only, not navigation accuracy

Time	Location on	Paired IMU in phase	Sensor age
10 s	False	2000	102.0 ms
20 s	False	2000	90.6 ms
30 s	False	2000	80.2 ms
40 s	False	2000	70.3 ms
50 s	False	1980	60.5 ms
90 s	False	8000	21.4 ms

A stationary USB-connected phone does not supply an independently measured moving route. Sensor continuity and GPS-withheld replay are separate checks.

Native position at checkpoints: 10 s: unavailable, 20 s: unavailable, 30 s: unavailable, 40 s: unavailable, 50 s: unavailable, 90 s: unavailable

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Build and user-flow checks

Check	Result	Evidence
APK identity	PASS	212,472,310 bytes; signature and 16 KiB alignment verified; installed app and website HTTP download match SHA-256.
Python regressions	PASS	265 passed, 2 skipped in 229.86 s. Skips require optional osmium. One ReportLab deprecation warning.
Android JVM tests	PASS	102 tests, zero failures/errors/skips. Unit tests and both APK builds succeed in 64 s.
Android lint	WARN	Zero errors, 25 existing warnings. Final separate lint invocation succeeds in 75 s; an earlier slow invocation was cancelled.
Latest phone inventory	PASS	84 discovered methods completed on the latest APK: 77 passed, 7 controlled/opt-in fixtures skipped, zero failures/errors. Each method uses a fresh instrumentation process without clearing user data. The six-duration replay runs separately with selected private rides.
User recording preservation	PASS	All 37 original SETULOG files remain byte-identical after the final complete device run. Only test-owned fixtures are removed.
Foreground screen wake	PASS	Production FLAG_KEEP_SCREEN_ON is asserted during recording and released after Stop. The 38.035-second physical foreground regression passes without a test-forced screen flag.
Normal-radio sensor capture	PASS	Latest 30-second foreground and 15-second background checks pass with Location enabled and Battery saver disabled. Maximum IMU gaps are 5.009 and 5.006 ms. These do not supersede the all-radios-off background failures.
Baseline focused phone tests	PASS	On the preceding aee9c570 build: 11 passed, 1 controlled-native-GNSS fixture skipped. Includes map rendering, settings, recording and curved native demo.

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Check	Result	Evidence
Baseline repeated journeys	PASS	Five demo/theme/settings/trips cycles in one process; baseline PSS 276,405-605,640 KiB; focused repeat 587,855-619,599 KiB. Both below unchanged 716,800 KiB cap.
Latest repeated journeys	PASS	Five journeys in the latest APK remain below the unchanged 716,800 KiB cap: PSS 609,280; 613,850; 612,414; 475,666; 608,178 KiB.
Compact map controls	PASS	The latest APK passes the new 8.665-second regression: all three map controls remain below the header and above attribution.
Native/model integrity	PASS	All nine packaged native-library and model files match the preceding APK byte-for-byte. The six-duration replay is repeated against the latest APK and yields the same counts.
Real Location-off continuity	PASS	90-second actual OS switch test; fresh paired IMU at all six checkpoints. No initialized native position or independent moving reference.
Beta website	PASS	Desktop and 360/390 px layouts, keyboard focus, reduced motion, denied clipboard, malformed release metadata, actual APK download and PDF HTTP checks. Local hosting only.
Background capture	FAIL	With Location and internet off, baseline background probes stall even with Battery saver off and battery exemptions enabled. A normal 45.71-second background recording has a 15.95-second IMU gap. Keep SETU visible; background capture is not certified.
90% within 10 m	FAIL	All six GPS-withheld replay durations fail the target. Availability itself is too low. This is development data, not independent field validation.

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Findings and remaining work

The original combined 81-test run has four retained failures: two Compose-root lifecycle failures, a 1,854,612 KiB map-memory reading against a 716,800 KiB ceiling, and an obsolete straight-chord assertion for the curved demo.

Use fresh instrumentation processes per method without clearing user data. Isolated focused tests and a five-cycle actual app soak pass; this supports fixture/process accumulation as a contributor, not a proof that every long app session is leak-free.

The curved-demo test now sums travelled distance along the path. Its travelled-distance and under-5-m synthetic error guards remain unchanged. No production estimator guard or model weight is relaxed.

Compact map controls were covered by attribution and initial loading feedback sat behind the task sheet. The latest APK reserves visible map space, lays compact controls out horizontally, and shows loading feedback in that visible area.

The latest APK keeps the screen on only during visible recording and releases that flag on stop/background. This is a foreground mitigation, not a claim that screen-off recording is reliable.

Background probes with Location, Wi-Fi and mobile data off stalled for 412,991 ms, and 155,377 ms with Battery saver disabled; foreground recovery was used for cleanup. A special-use service candidate also failed and was reverted. The normal app independently shows a roughly 15.95-second sensor gap in a 45.71-second background interval. These failures remain disclosed with their source APKs.

Replay now scores exactly duration-times-10 ticks, excludes the GPS recovery boundary and compares position with the reference at the scoring timestamp. Missing predictions count as failures; missing references remain unknown, not successes.

The HTML regression test permits only the intentional trusted sweep script and rejects route-name script injection. The beta publisher rejects mismatched evidence, retains the preceding release after failure and optionally watches verified artifacts every 30 seconds.

Three same-phone/day scooter rides cannot establish independent accuracy. Repeated ticks are correlated. Conditional low errors on rare surviving estimates are not whole-outage accuracy. All matched ninety-second-cohort predictions cease after ten seconds.

Sustained GPS-free navigation and heading initialization remain release blockers. The earlier CPU-trained model still predicts motion at rest and remains research-only. Collect rigid-mount, multi-day rides with independent position reference before making accuracy claims.

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Device checks

baseline-all-device-tests.log

Suite APK SHA-256: aee9c5706879a44790b62d4347df75e6bbe3366e08c1d564e6d4fa437a208d5b

Runner completed: True. Counts: {'skipped': 7, 'failed': 4, 'passed': 70}

Test	Result
AndroidWorkflowTest.liveNativePositioningConsumesAndroidSensorsAndRecordsSeparateOutput	skipped
AndroidWorkflowTest.offlineJourneyAndSettingsRemainUsable	failed
AndroidWorkflowTest.presentationDemoShowsNativeLossRecoveryAndSurvivesRecreation	failed
AndroidWorkflowTest.liveTrackingFollowsMockFixesRecordsAndSavesWithoutADestination	passed
AndroidWorkflowTest.missingMeasuredAndStaleGpsStayDistinct	passed
AndroidWorkflowTest.sensorSubscriptionsCanBeRepeatedlyStoppedAndRestarted	passed
AndroidWorkflowTest.regionalMapsInstallSwitchRouteRejectAndRemove	passed
CompiledRoadGraphTest.pythonCompiledFixtureMatchesJsonRoutesAndOneWaySemantics	passed
CompiledRoadGraphTest.ncrColdAndWarmRoutesUseMappedGraphAndRecoverCorruptCacheWithoutJsonParsing	passed
GnssObservationTest.duplicateDelayedAndFutureFixesDoNotReplaceLatestOrEnterLog	passed
GnssObservationTest.malformedMeasurementMetadataIsRejected	passed
GnssObservationTest.invalidOrOrphanMeasurementsCannotBecomeObservations	passed
GnssObservationTest.optionalMeasurementsPreserveMissingAndMeasuredZero	passed

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Test	Result
GnssObservationTest.legacyZeroMeasurementsAreUnknownAndNonzeroValuesSurvive	passed
LocationLifecycleTest.coldStartWithLocationOffObservesReenableWithoutRestart	passed
MapPackStoreTest.invalidGeometryMetadataAndGraphAreRejected	passed
MapPackStoreTest.includedNcrUsesBoundedLocalVectorTiles	passed
MapPackStoreTest.corruptInstalledMapCannotActivateAndRestoresFallback	passed
MapPackStoreTest.installActivateRestoreAndRemovePreserveFallback	passed
MapPackStoreTest.includedNcrCoversCitiesAndRoutesShahdaraToMait	passed
MapPackStoreTest.revisionsNeverSilentlyReplaceOrDowngrade	passed
MapPackStoreTest.directDownloadChecksHashAndDoesNotActivateAutomatically	passed
MapPackStoreTest.badArchivesAndChecksumsLeaveNoInstalledFiles	passed
MapPackStoreTest.cancellationKeepsThePreviousMapAndCleansStaging	passed
MapPackStoreTest.failedDownloadsNeverInstallOrFollowRedirects	passed
MapRenderingTest.ncrTilesRenderAcrossZoomsAndRepeatedMapDisposalWithoutHeapGrowth	failed
MapRenderingTest.plannedNcrRouteRemainsRenderedWhenFollowingAndChangingTheme	passed
ModelConsentTest.legacyFlagDoesNotAuthorizeUploadsAndChangingServersRevokesConsent	passed
ModelProviderTest.unavailableRetainsReasonRatherThanPretendingTheVehicleStopped	passed
ModelProviderTest.malformedAndOutOfWindowMeasurementsAreRejected	passed

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Test	Result
ModelProviderTest.cancellingADownloadDoesNotWaitForTheReadTimeout	passed
ModelProviderTest.healthPreservesExperimentalWarningAndInferenceContainsNoLocation	passed
ModelProviderTest.urlAndWindowValidationHappenBeforeNetworkTraffic	passed
ModelProviderTest.redirectsOversizedResponsesAndUnknownHealthAreRejected	passed
NativeEngineTest.stationaryUncalibratedVehicleDoesNotRunIndefinitely	passed
NativeEngineTest.horizontalVelocityUpdateDoesNotInventVerticalSpeed	passed
NativeEngineTest.rapidPhoneRotationClearsVehicleMountAndRequiresFreshAlignment	passed
NativeEngineTest.magneticDeclinationMatchesPublishedNoaaWmm2025Examples	passed
NativeEngineTest.pendingFixDoesNotResetTheFilterAndOutageEventuallyWithholdsOutput	passed
NativeEngineTest.measuredStopWithoutCourseUsesItsSpeedUncertainty	passed
NativeEngineTest.delayedCorrectionsMatchOnTimePropagationAndGeographicDisplacement	passed
NativeEngineTest.gapsWithholdTheEstimateAndRequireNewAlignmentInputs	passed
NativeEngineTest.nonCarProfileKeepsRawRotationButNeverBorrowsCarConstraintsOrOutageBudget	passed
NativeEngineTest.learnedSpeedRejectsFutureDuplicateAndStaleMeasurements	passed
NativeEngineTest.initializationNeedsHeadingAndMissingAltitudeDoesNotBecomeSeaLevel	passed
NativeEngineTest.gatesAndInvalidTimestampsDoNotMoveTheState	passed
NativeFilterTest.matchesPythonAcrossPropagationAndEveryMeasurementChannel	passed

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Test	Result
NativeFilterTest.invalidSamplesDoNotCorruptStateAndClosedHandlesAreSafe	passed
OfflineMapTest.outOfCoverageOriginIsRejected	passed
OfflineMapTest.includedDestinationsHaveDrivableRoutes	passed
OfflineTilesTest.missingTruncatedAndSameSizeCorruptTilesAreRepairedDespiteACompleteMarker	passed
OnDeviceModelProviderTest.actualPackagedModelRunsLocallyButUnapprovedPredictionsCannotFuse	passed
PhysicalRecordingTest.actualImuRecordingContinuesInForegroundWithoutLocationOrModelSharing	passed
PhysicalRecordingTest.actualImuRecordingContinuesInBackgroundWithoutLocationOrModelSharing	passed
PositioningDemoTest.nativeEngineTracksThroughWithheldGpsAndAcceptsRecovery	failed
PositioningDemoTest.cancelledPreparationDoesNotContinueGeneratingFrames	passed
ScooterCaptureReviewTest.changingCarToScooterReconfiguresSensorsAndDoesNotStartCarModel	passed
ScooterCaptureReviewTest.exportSelectedRidesAndMeasureIndependentBlackouts	skipped
ScooterCaptureReviewTest.measureLongWarmupBlackoutsWithoutChangingRecordings	skipped
SensorFallbackTest.missingHardwareHeadingAccuracyStillUsesRealSensorPipelineThroughTurns	passed
SensorFallbackTest.capturedPhysicalSensorsCanBeReplayedWithoutLeakingCoordinates	skipped
SensorFallbackTest.nativePositionsPublishAtTenHertzDuringGpsAndSensorOnlySegments	passed

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Test	Result
SensorFallbackTest.gpsMotionCalibratesThroughAccelerationAndTurnsWithoutACompassHeading	passed
SensorFallbackTest.coldStartWithoutGpsNeverInventsAnAbsolutePosition	passed
SensorFallbackTest.disturbedMagnetometerDoesNotBypassHeadingChecks	passed
TrainingCollectionTest.phoneRecordsAndExportsPortableTrainingBundle	passed
TrainingCollectionTest.gpsReferenceChangesCannotLeakIntoTheWithheldTrajectory	passed
TrainingCollectionTest.shadowReplayWorksAcrossRebootsAndReportsMissingHeadingHonestly	passed
TrainingCollectionUiTest.trainingControlsExplainCollectionAndRequireExplicitExport	passed
TripStoreTest.malformedImportsLeaveNoOrphanFiles	passed
TripStoreTest.importExportAndDeleteRoundTrip	passed
TripStoreTest.recordedTrajectoryRetainsInertialPointsAndUncertaintySeparatelyFromRawGps	passed
TripStoreTest.invalidTrajectoryUncertaintyAndUnknownStreamsAreRejected	passed
TripStoreTest.optionalMeasurementsSurviveImportAndStoredMetadata	passed
TripStoreTest.activeRecordingIsNotRecovered	passed
TripStoreTest.interruptedFinalRecordIsRecoveredAndExportsCleanly	passed
WalkingIntegrationTest.physicalWalkingProfileRecordsImuWithoutRunningCarModel	passed
WalkingIntegrationTest.physicalWalkingEstimateSurvivesGpsOffAtRest	skipped
WalkingIntegrationTest.replayCapturedStepBatchesWithRecordedInitialHeading	skipped

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Test	Result
WalkingIntegrationTest.syntheticPlatformCallbacksUseStepsAndWithholdWithoutPermission	passed
WalkingIntegrationTest.replayPrivateWalkingRecordingsChecksTheUncalibratedCeiling	skipped

AndroidWorkflowTest.offlineJourneyAndSettingsRemainUsable

java.lang.IllegalStateException: No compose hierarchies found in the app. Possible reasons include: (1) the Activity that calls setContent did not launch; (2) setContent was not called; (3) setContent was called before the ComposeTestRule ran. If setContent is called by the Activity, make sure the Activity is launched after the ComposeTestRule runs

AndroidWorkflowTest.presentationDemoShowsNativeLossRecoveryAndSurvivesRecreation

java.lang.IllegalStateException: No compose hierarchies found in the app. Possible reasons include: (1) the Activity that calls setContent did not launch; (2) setContent was not called; (3) setContent was called before the ComposeTestRule ran. If setContent is called by the Activity, make sure the Activity is launched after the ComposeTestRule runs

MapRenderingTest.ncrTilesRenderAcrossZoomsAndRepeatedMapDisposalWithoutHeapGrowth

java.lang.AssertionError: Map renderer memory is not bounded: 1854612 KiB

PositioningDemoTest.nativeEngineTracksThroughWithheldGpsAndAcceptsRecovery

java.lang.AssertionError

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Device checks

actual-location-off.log

Suite APK SHA-256: aee9c5706879a44790b62d4347df75e6bbe3366e08c1d564e6d4fa437a208d5b

Runner completed: True. Counts: {'passed': 1}

Test	Result
LocationOffDurationTest.realSensorsContinueForNinetySecondsWithSystemLocationOff	passed

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Device checks

latest-six-duration-replay.log

Suite APK SHA-256: 410ca8d79d42780b3cb1c33c0d569c093e1216b257d9c2ccd10dc56e9a8db62b

Runner completed: True. Counts: {'passed': 1}

Test	Result
ScooterCaptureReviewTest.measureLongWarmupBlackoutsWithoutChangingRecordings	passed

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Device checks

internet-location-off-recording.log

Suite APK SHA-256: aee9c5706879a44790b62d4347df75e6bbe3366e08c1d564e6d4fa437a208d5b

Runner completed: True. Counts: {'passed': 1, 'failed': 1}

Test	Result
PhysicalRecordingTest.actualImuRecordingContinuesInForegroundWithoutLocationOrModelSharing	passed
PhysicalRecordingTest.actualImuRecordingContinuesInBackgroundWithoutLocationOrModelSharing	failed

PhysicalRecordingTest.actualImuRecordingContinuesInBackgroundWithoutLocationOrModelSharing

java.lang.AssertionError: Process was suspended for 412991 ms

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Device checks

internet-location-off-no-saver.log

Suite APK SHA-256: aee9c5706879a44790b62d4347df75e6bbe3366e08c1d564e6d4fa437a208d5b

Runner completed: True. Counts: {'passed': 1, 'failed': 1}

Test	Result
PhysicalRecordingTest.actualImuRecordingContinuesInForegroundWithoutLocationOrModelSharing	passed
PhysicalRecordingTest.actualImuRecordingContinuesInBackgroundWithoutLocationOrModelSharing	failed

PhysicalRecordingTest.actualImuRecordingContinuesInBackgroundWithoutLocationOrModelSharing

java.lang.AssertionError: Process was suspended for 155377 ms

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Device checks

special-use-background.log

Suite APK SHA-256: bf4ec19ae8b6aff661e9dc5653373f093310e046de77d0399163cd1bf33f0475

Runner completed: True. Counts: {'failed': 1}

Test	Result
PhysicalRecordingTest.actualImuRecordingContinuesInBackgroundWithoutLocationOrModelSharing	failed

PhysicalRecordingTest.actualImuRecordingContinuesInBackgroundWithoutLocationOrModelSharing

java.lang.AssertionError: Process was suspended for 159940 ms

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Device checks

Isolated device test methods

Suite APK SHA-256: 410ca8d79d42780b3cb1c33c0d569c093e1216b257d9c2ccd10dc56e9a8db62b

Runner completed: True. Counts: {'skipped': 7, 'passed': 77}

Test	Result
AndroidWorkflowTest.liveNativePositioningConsumesAndroidSensorsAndRecordsSeparateOutput	skipped
AndroidWorkflowTest.offlineJourneyAndSettingsRemainUsable	passed
AndroidWorkflowTest.presentationDemoShowsNativeLossRecoveryAndSurvivesRecreation	passed
AndroidWorkflowTest.liveTrackingFollowsMockFixesRecordsAndSavesWithoutADestination	passed
AndroidWorkflowTest.missingMeasuredAndStaleGpsStayDistinct	passed
AndroidWorkflowTest.sensorSubscriptionsCanBeRepeatedlyStoppedAndRestarted	passed
AndroidWorkflowTest.regionalMapsInstallSwitchRouteRejectAndRemove	passed
AndroidWorkflowTest.repeatedDemoAndThemeJourneysKeepMemoryBounded	passed
AndroidWorkflowTest.compactMapControlsStayAboveTheDriveSheet	passed
CompiledRoadGraphTest.pythonCompiledFixtureMatchesJsonRoutesAndOneWaySemantics	passed
CompiledRoadGraphTest.ncrColdAndWarmRoutesUseMappedGraphAndRecoverCorruptCacheWithoutJsonParsing	passed
GnssObservationTest.duplicateDelayedAndFutureFixesDoNotReplaceLatestOrEnterLog	passed
GnssObservationTest.malformedMeasurementMetadatalRejected	passed

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Test	Result
GnssObservationTest.invalidOrOrphanMeasurementsCannotBecomeObservations	passed
GnssObservationTest.optionalMeasurementsPreserveMissingAndMeasuredZero	passed
GnssObservationTest.legacyZeroMeasurementsAreUnknownAndNonzeroValuesSurvive	passed
LocationLifecycleTest.coldStartWithLocationOffObservesReenableWithoutRestart	passed
LocationOffDurationTest.realSensorsContinueForNinetySecondsWithSystemLocationOff	passed
MapPackStoreTest.invalidGeometryMetadataAndGraphAreRejected	passed
MapPackStoreTest.includedNcrUsesBoundedLocalVectorTiles	passed
MapPackStoreTest.corruptInstalledMapCannotActivateAndRestoresFallback	passed
MapPackStoreTest.installActivateRestoreAndRemovePreserveFallback	passed
MapPackStoreTest.includedNcrCoversCitiesAndRoutesShahdaraToMait	passed
MapPackStoreTest.revisionsNeverSilentlyReplaceOrDowngrade	passed
MapPackStoreTest.directDownloadChecksHashAndDoesNotActivateAutomatically	passed
MapPackStoreTest.badArchivesAndChecksumsLeaveNoInstalledFiles	passed
MapPackStoreTest.cancellationKeepsThePreviousMapAndCleansStaging	passed
MapPackStoreTest.failedDownloadsNeverInstallOrFollowRedirects	passed
MapRenderingTest.ncrTilesRenderAcrossZoomsAndRepeatedMapDisposalWithoutHeapGrowth	passed
MapRenderingTest.plannedNcrRouteRemainsRenderedWhenFollowingAndChangingTheme	passed

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Test	Result
ModelConsentTest.legacyFlagDoesNotAuthorizeUploadsAndChangingServersRevokesConsent	passed
ModelProviderTest.unavailableRetainsReasonRatherThanPretendingTheVehicleStopped	passed
ModelProviderTest.malformedAndOutOfWindowMeasurementsAreRejected	passed
ModelProviderTest.cancellingADownloadDoesNotWaitForTheReadTimeout	passed
ModelProviderTest.healthPreservesExperimentalWarningAndInferenceContainsNoLocation	passed
ModelProviderTest.urlAndWindowValidationHappenBeforeNetworkTraffic	passed
ModelProviderTest.redirectsOversizedResponsesAndUnknownHealthAreRejected	passed
NativeEngineTest.stationaryUncalibratedVehicleDoesNotRunIndefinitely	passed
NativeEngineTest.horizontalVelocityUpdateDoesNotInventVerticalSpeed	passed
NativeEngineTest.rapidPhoneRotationClearsVehicleMountAndRequiresFreshAlignment	passed
NativeEngineTest.magneticDeclinationMatchesPublishedNoaaWmm2025Examples	passed
NativeEngineTest.pendingFixDoesNotResetTheFilterAndOutageEventuallyWithholdsOutput	passed
NativeEngineTest.measuredStopWithoutCourseUsesItsSpeedUncertainty	passed
NativeEngineTest.delayedCorrectionsMatchOnTimePropagationAndGeographicDisplacement	passed
NativeEngineTest.gapsWithholdTheEstimateAndRequireNewAlignmentInputs	passed
NativeEngineTest.nonCarProfileKeepsRawRotationButNeverBorrowsCarConstraintsOrOutageBudget	passed
NativeEngineTest.learnedSpeedRejectsFutureDuplicateAndStaleMeasurements	passed

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Test	Result
NativeEngineTest.initializationNeedsHeadingAndMissingAltitudeDoesNotBecomeSeaLevel	passed
NativeEngineTest.gatesAndInvalidTimestampsDoNotMoveTheState	passed
NativeFilterTest.matchesPythonAcrossPropagationAndEveryMeasurementChannel	passed
NativeFilterTest.invalidSamplesDoNotCorruptStateAndClosedHandlesAreSafe	passed
OfflineMapTest.outOfCoverageOriginIsRejected	passed
OfflineMapTest.includedDestinationsHaveDrivableRoutes	passed
OfflineTilesTest.missingTruncatedAndSameSizeCorruptTilesAreRepairedDespiteACompleteMarker	passed
OnDeviceModelProviderTest.actualPackagedModelRunsLocallyButUnapprovedPredictionsCannotFuse	passed
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PhysicalRecordingTest.actualImuRecordingContinuesInBackgroundWithoutLocationOrModelSharing	passed
PositioningDemoTest.nativeEngineTracksThroughWithheldGpsAndAcceptsRecovery	passed
PositioningDemoTest.cancelledPreparationDoesNotContinueGeneratingFrames	passed
ScooterCaptureReviewTest.changingCarToScooterReconfiguresSensorsAndDoesNotStartCarModel	passed
ScooterCaptureReviewTest.exportSelectedRidesAndMeasureIndependentBlackouts	skipped
ScooterCaptureReviewTest.measureLongWarmupBlackoutsWithoutChangingRecordings	skipped

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Test	Result
SensorFallbackTest.missingHardwareHeadingAccuracyStillUsesRealSensorPipelineThroughTurns	passed
SensorFallbackTest.capturedPhysicalSensorsCanBeReplayedWithoutLeakingCoordinates	skipped
SensorFallbackTest.nativePositionsPublishAtTenHertzDuringGpsAndSensorOnlySegments	passed
SensorFallbackTest.gpsMotionCalibratesThroughAccelerationAndTurnsWithoutACompassHeading	passed
SensorFallbackTest.coldStartWithoutGpsNeverInventsAnAbsolutePosition	passed
SensorFallbackTest.disturbedMagnetometerDoesNotBypassHeadingChecks	passed
TrainingCollectionTest.phoneRecordsAndExportsPortableTrainingBundle	passed
TrainingCollectionTest.gpsReferenceChangesCannotLeakIntoTheWithheldTrajectory	passed
TrainingCollectionTest.shadowReplayWorksAcrossRebootsAndReportsMissingHeadingHonestly	passed
TrainingCollectionUiTest.trainingControlsExplainCollectionAndRequireExplicitExport	passed
TripStoreTest.malformedImportsLeaveNoOrphanFiles	passed
TripStoreTest.importExportAndDeleteRoundTrip	passed
TripStoreTest.recordedTrajectoryRetainsInertialPointsAndUncertaintySeparatelyFromRawGps	passed
TripStoreTest.invalidTrajectoryUncertaintyAndUnknownStreamsAreRejected	passed
TripStoreTest.optionalMeasurementsSurviveImportAndStoredMetadata	passed
TripStoreTest.activeRecordingIsNotRecovered	passed
TripStoreTest.interruptedFinalRecordsRecoveredAndExportsCleanly	passed

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Test	Result
WalkingIntegrationTest.physicalWalkingProfileRecordsImuWithoutRunningCarModel	passed
WalkingIntegrationTest.physicalWalkingEstimateSurvivesGpsOffAtRest	skipped
WalkingIntegrationTest.replayCapturedStepBatchesWithRecordedInitialHeading	skipped
WalkingIntegrationTest.syntheticPlatformCallbacksUseStepsAndWithholdWithoutPermission	passed
WalkingIntegrationTest.replayPrivateWalkingRecordingsChecksTheUncalibratedCeiling	skipped